

Exp. 6
Date: 24/08/24

Simulation of Mean, variance and cross correlation using scilab

AIM:

To calculate mean and variance of a given data set using scilab.

APPARATUS REQUIRED:

- * Computer / Laptop.
- * Scilab software

CODE:

```
clc;  
clear;  
x = [5 8 9 10 11];  
N = length(x);  
mean_value = sum(x) / N;  
variance_value = sum((x - mean_value).^2) / N;  
disp("mean = ");  
disp(mean_value);  
disp("variance = ");  
disp(variance_value);
```

Calculation

Given

$$\text{mean} = \frac{\sum x}{n}$$

Add all numbers:

$$5 + 8 + 9 + 10 + 11 = 43$$

$$\text{mean} = \frac{43}{5} = 8.6$$

variance:

$$\sigma^2 = \frac{\sum (x - \mu)^2}{n}$$

mean $\mu = 8.666$

x	$x - \mu$	$(x - \mu)^2$
5	$5 - 8.666 = -3.666$	13.435
8	$8 - 8.666 = -0.666$	0.4435
9	$9 - 8.666 = 0.333$	0.111
10	$10 - 8.666 = 1.333$	1.7775
11	$11 - 8.666 = 2.333$	5.4435
$\sum$		22.2

x	$x - \mu$	$(x - \mu)^2$
5	$5 - 8.6 = -3.6$	12.96
8	$8 - 8.6 = -0.6$	0.36
9	$9 - 8.6 = 0.4$	0.16
10	$10 - 8.6 = 1.4$	1.96
11	$11 - 8.6 = 2.4$	5.76
$\sum$		21.16

Calculation :

$$\cancel{14.2008} + \cancel{14.2008} + \cancel{0.4435} + \cancel{2.7756} + \cancel{7.1075}$$
$$= \frac{\cancel{38.2281}}{5}$$

$$\text{variance} = 7.7456$$

$$12.96 + 12.96 + 0.36 + 2.56 + 6.76$$

$$= \frac{35.6}{5}$$

$$= 7.12$$

OUTPUT.

$$\text{Mean} = 8.6$$

$$\text{variance} = 7.12$$

O/P

Marks allocation :

Scilab program implementation (20)	Work form and lab report (20)	Observation and result (20)	Viva - voce (20)	Record observation (20)	Total marks
20	20	20	18	20	96

RESULT:

The mean and variance of the given data set were successfully calculated using scilab

* Mean = 8.6

* Variance = 7.12